LEC 4.2: SIGNALING THROUGH GPCR

Learning goals:

- To understand how GPCRs signal in the cell
- To know the different types of GPCR effectors and secondary messengers
- To learn examples of normal physiological and pathological processes related to GPCRs

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Biological relevance of G Protein-Coupled Receptors (GPCRs)

Largest family of receptors (>1000 genes in the human genome)

- **Role in sensory organs**: Vision, olfaction, taste

- Receptors for neurotransmitters

Acetylcholine: heart rate decrease, muscle contraction

Serotonin: regulation of mood, sleep and digestion

Dopamine: motivation, intention, memory, learning, and control of body movements

Histamine: allergies, digestion, wake-sleep cycle

- Receptors of hormones

Glucagon: glucose release

Adrenaline: response to danger (fight or flight)

- Receptors of growth factors

Robert J. Lefkowitz

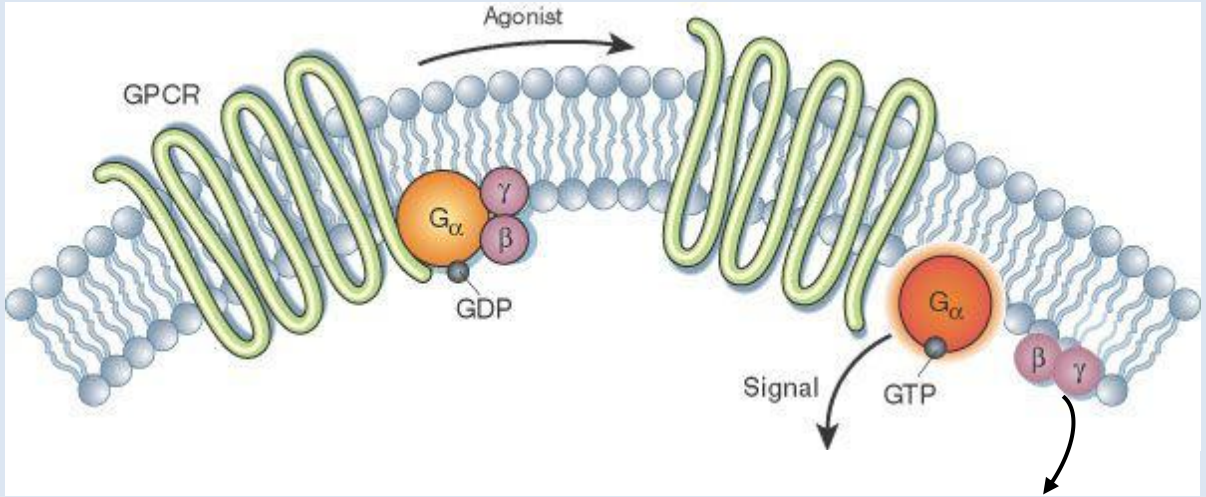


Brian K. Kobilka



Nobel Price in Chemistry 2012

Elements of GPCRs



- ***Receptor***:

  7-pass transmembrane receptors

- ***G protein***:

  Heterotrimeric (α, β and γ subunits)

  Connecting membrane-anchored protein

  Bind Guanine nucleotides (GTP, GDP)

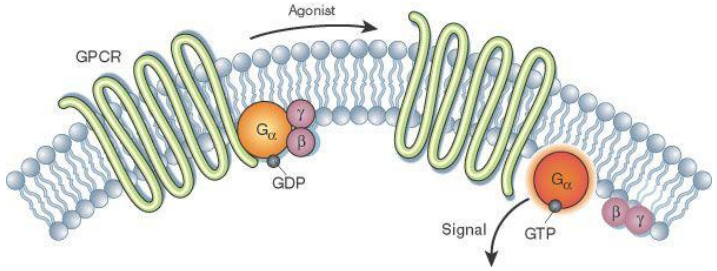
  GDP-bound: inactive

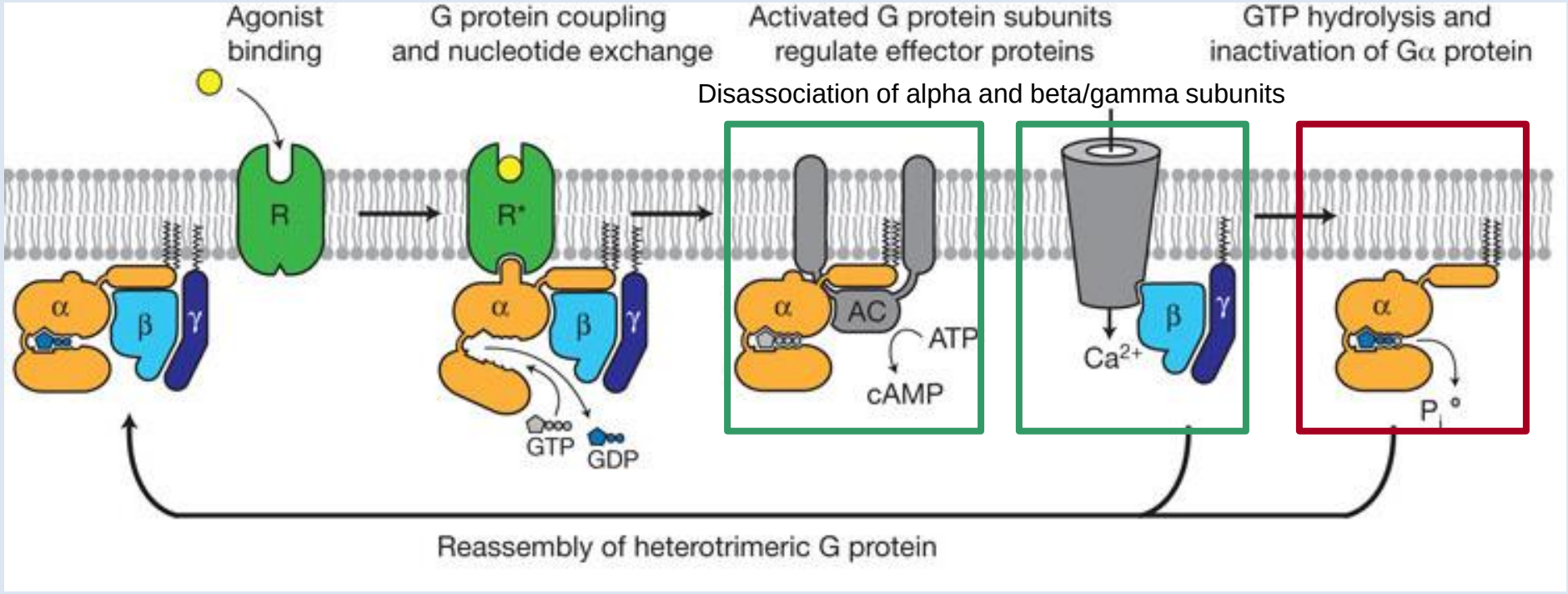
  GTP-bound: active dissociate in α-GTP and $\beta\gamma$

- ***Enzyme or ion channel***:

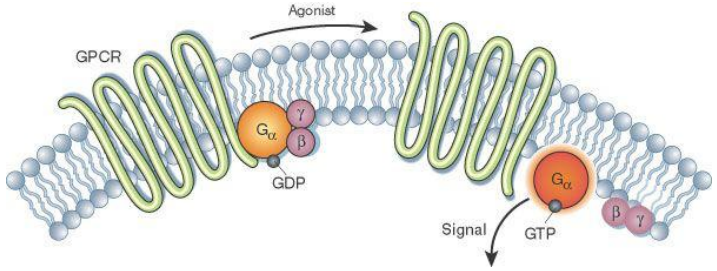
  It produces (e.g., cAMP) or increases (e.g., $Ca2^{+}$) a second messenger

Elements of GPCRs





G-Proteins



TYPES OF G PROTEINS:

- **Gs (stimulatory):** **Induce cAMP production via adenylate cyclase**

- **Gi (inhibitory):** **Inhibit cAMP synthesis and change ion fluxes**

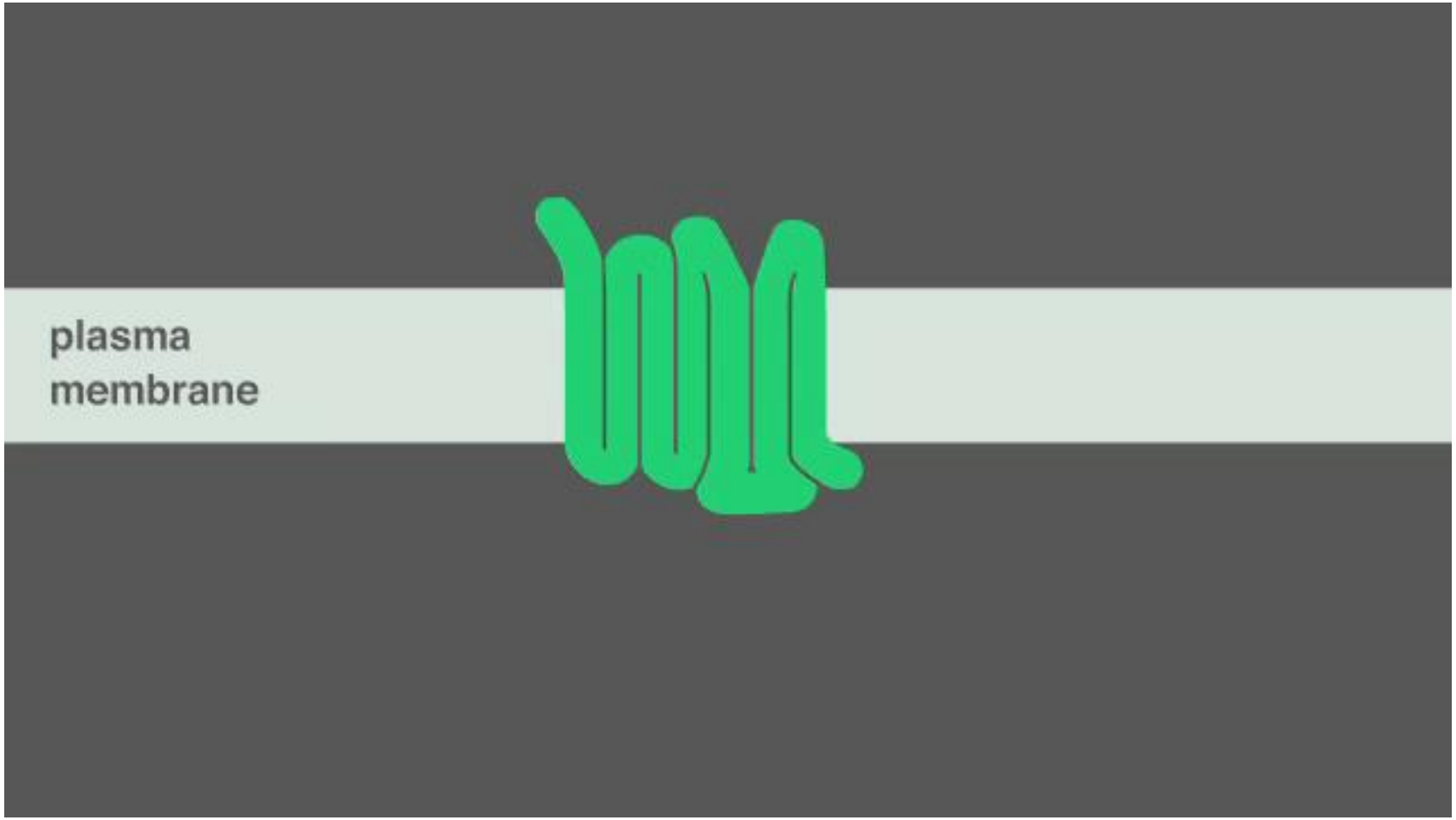
Adrenaline

1. Bind to β-adrenergic receptor in hepatocytes
2. Gα subunit functions as Gs
3. Induces cAMP
4. Enzymes increase glucose in blood

1. Bind to α-adrenergic receptor in smooth muscle
2. Gα subunit functions as Gi
3. Inhibits cAMP
4. Muscle relaxation

Video on how GPCR signaling works

https://www.youtube.com/watch?v=B-fJat9__G0



Features of GPCR signaling

1. Follow same general signaling features: ligand-receptor signaling, ON/OFF balance, integration of signals, signal amplification, and one molecule-different responses.

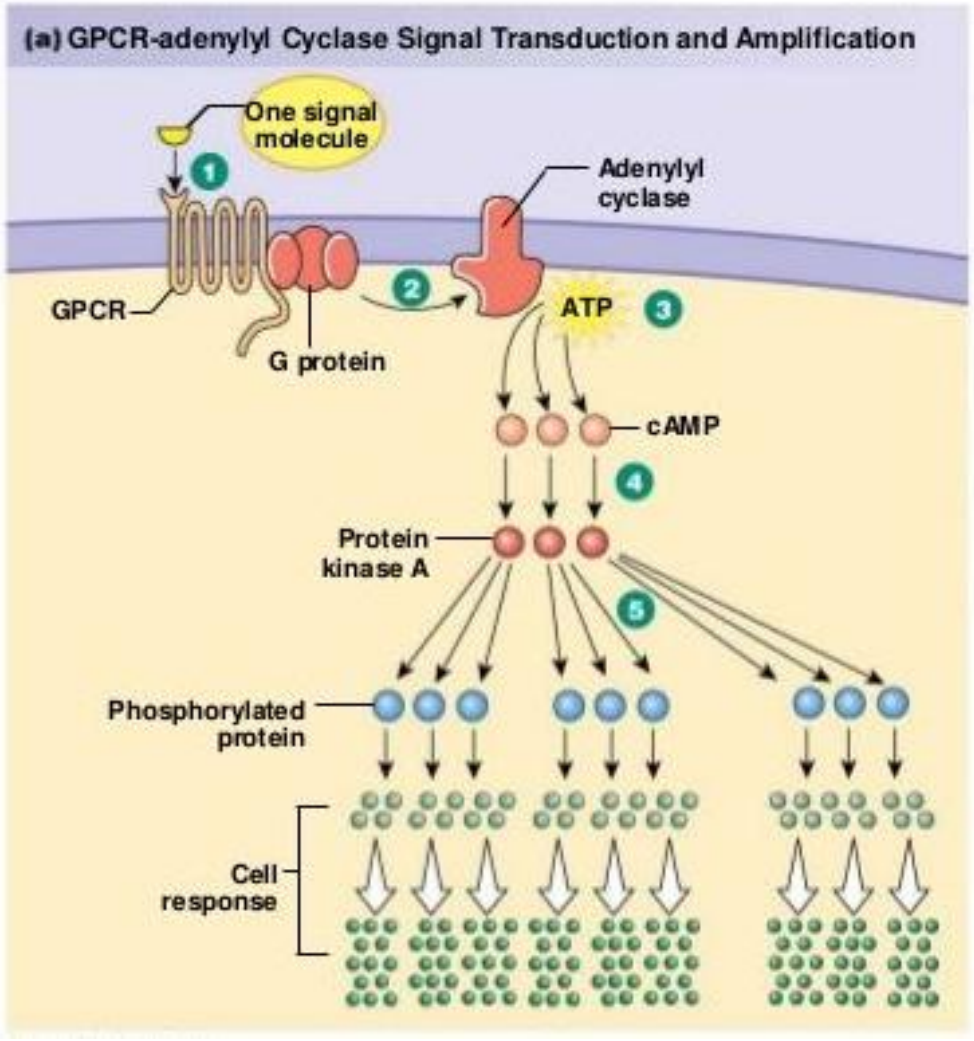
2. Structurally similar receptors release different signals

3. A few effectors engage a plethora of signaling pathways to regulate multiple cellular functions almost in any human tissue

4. Exquisite sensitivity to signaling molecules/ligands

5. From the point of view of pharmacological inhibition: Difficulties to design specific inhibitors to GPCRs

GPCR features: Signal amplification process



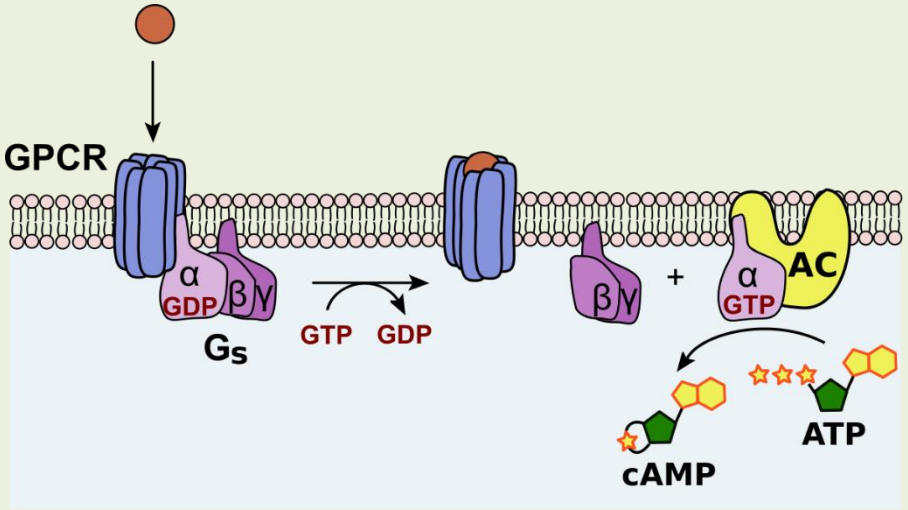
(a) Signaling pathway	(b) Number of molecules activated
RECEPTION Binding of epinephrine to G protein-linked receptor	1 molecule
TRANSDUCTION Inactive G protein → Active G protein	10^2 molecules
Inactive adenylyl cyclase → Active adenylyl cyclase	10^2 molecules
ATP → Cyclic AMP	10^4 molecules
Inactive protein kinase A → Active protein kinase A	10^4 molecules
Inactive phosphorylase kinase → Active phosphorylase kinase	10^5 molecules
Inactive glycogen phosphorylase → Active glycogen phosphorylase	10^6 molecules
RESPONSE Glycogen → Glucose-1-phosphate	10^8 molecules

GPCR effectors: Adenylate cyclase

Anchored to plasma membrane

Activated by Gα-GTP

- cAMP production (from ATP): second messenger
- cAMP excess eliminated by phosphodiesterase



Effects of cAMP

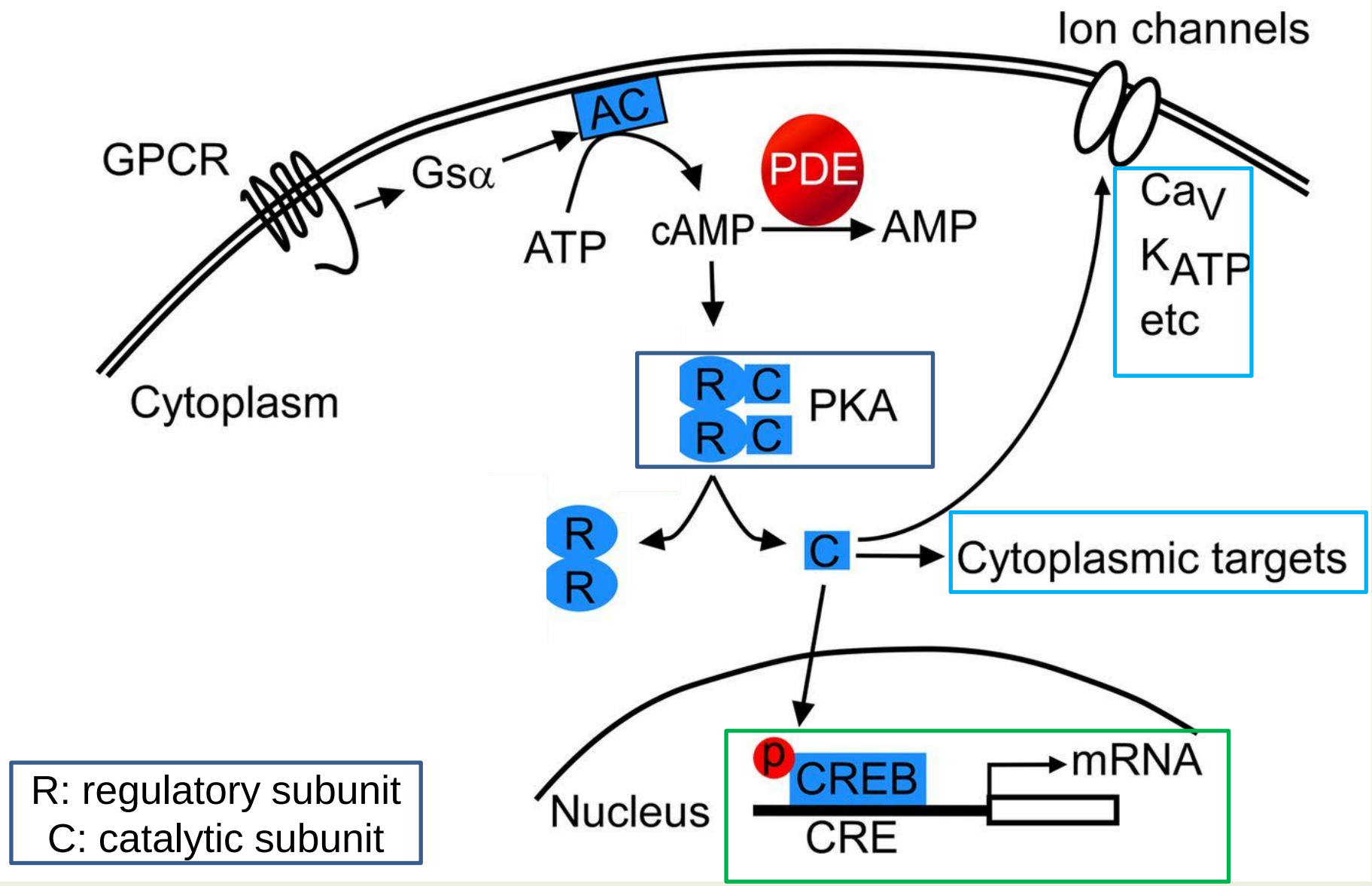
*Fast response: Does not require gene expression

Smell sense (1 cell-1 receptor type): ion channels (Na+) → mbr. depolarization

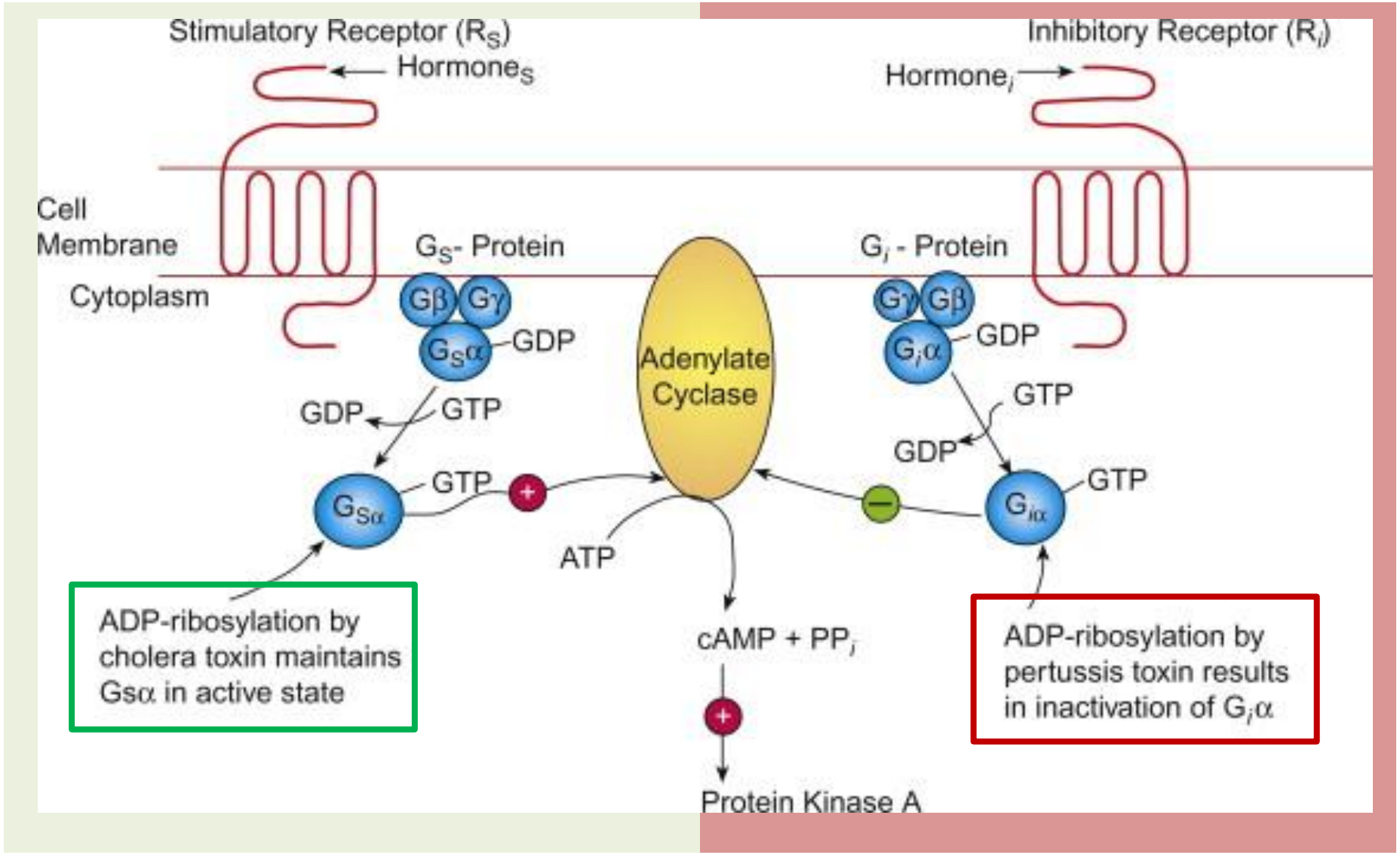
*Slower response: Binding and activating protein kinase A (in most cells)

Glycogenolysis in muscle/liver

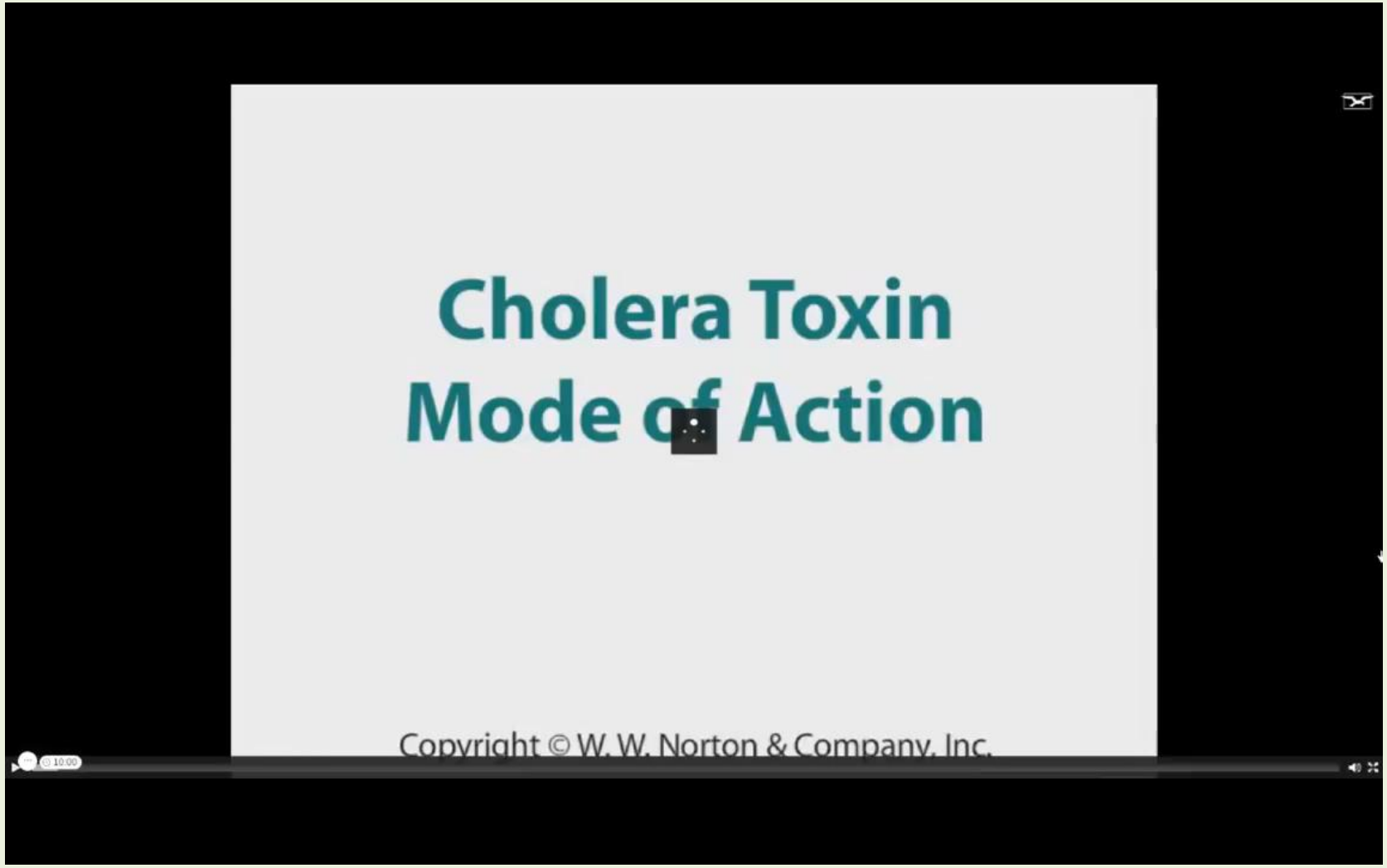
GPCR effectors: Adenylate cyclase



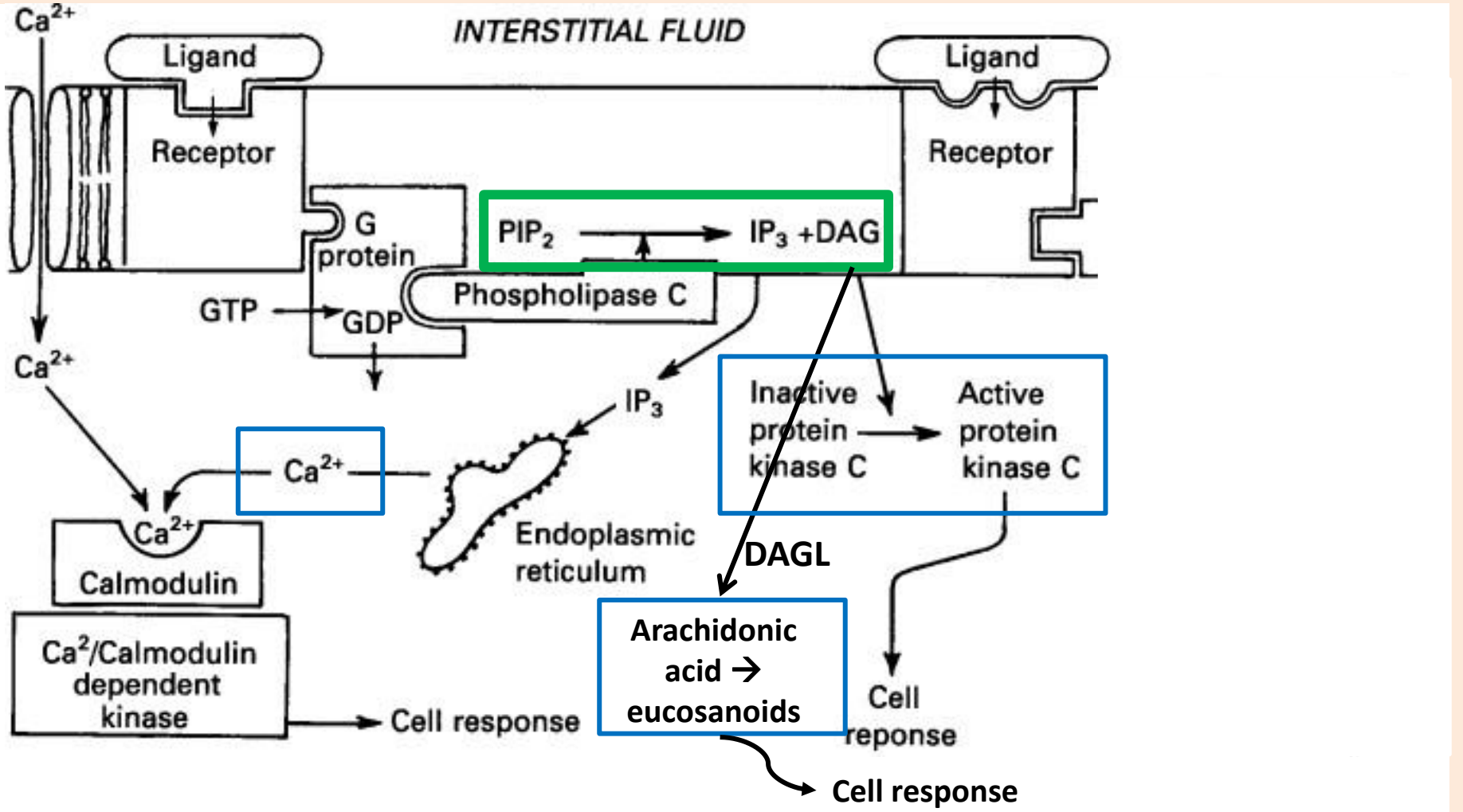
Alteration of GPCR in the intestine and respiratory system by Cholera and Bordetella Pertussis toxins



Alteration of GPCR in the intestine by cholera toxin: the importance of shutting off signaling



GPCR effectors: Phospholipase C-β



PIP2: phosphatidylinositol-4,5 bisphosphate; **DAG:** diacylglycerol; **DAGL:** diacylglycerol lipase
IP_3: inositol triphosphate; **PKC:** Protein Kinase C

GPCR effectors: Phosphodiesterase

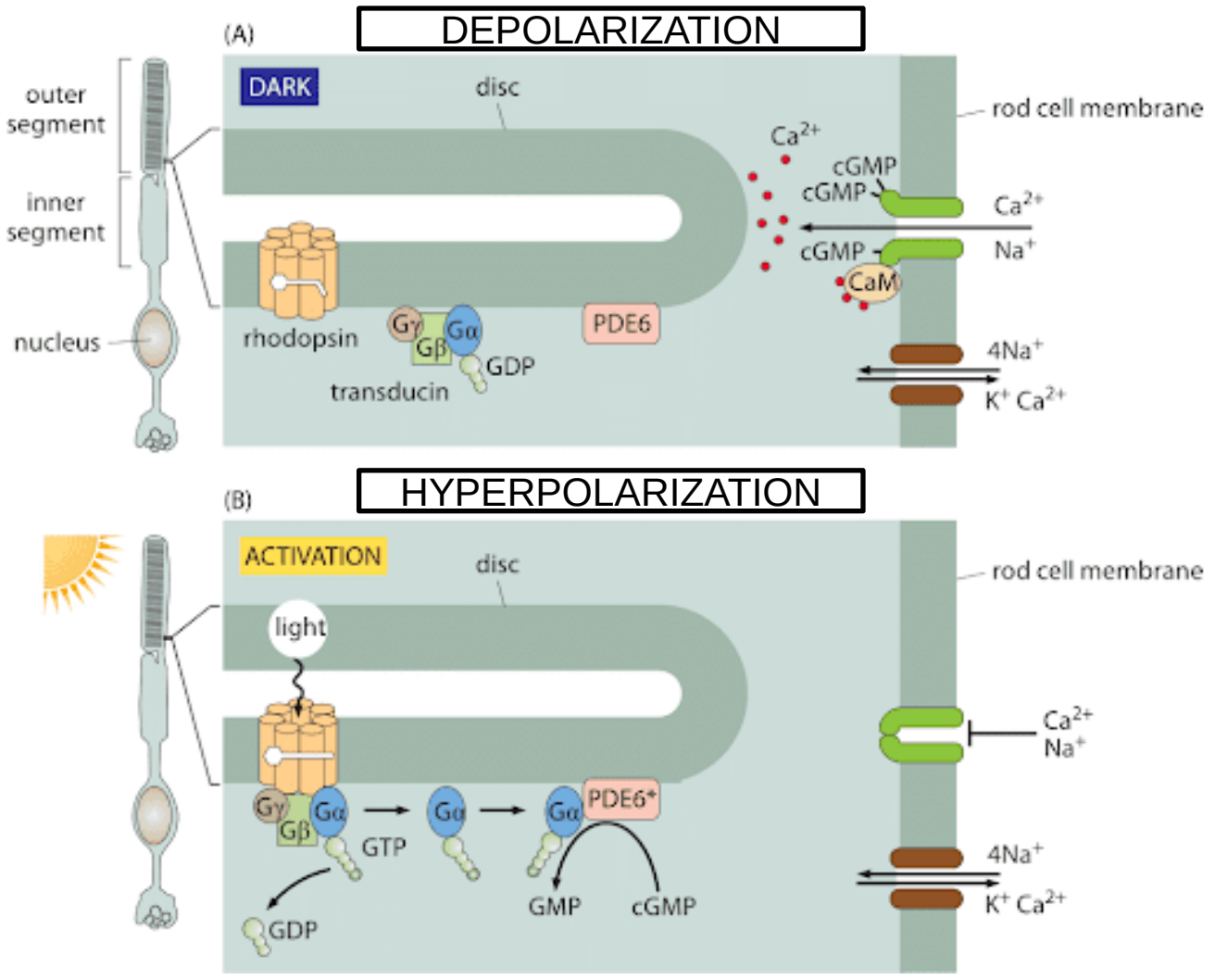
Converts cGMP to GMP to balance its intracellular levels

cGMP is produced by Guanylate Cyclase (GC) (e.g., NO signaling in blood vessels)

GPCRs IN SIGHT SIGNALING

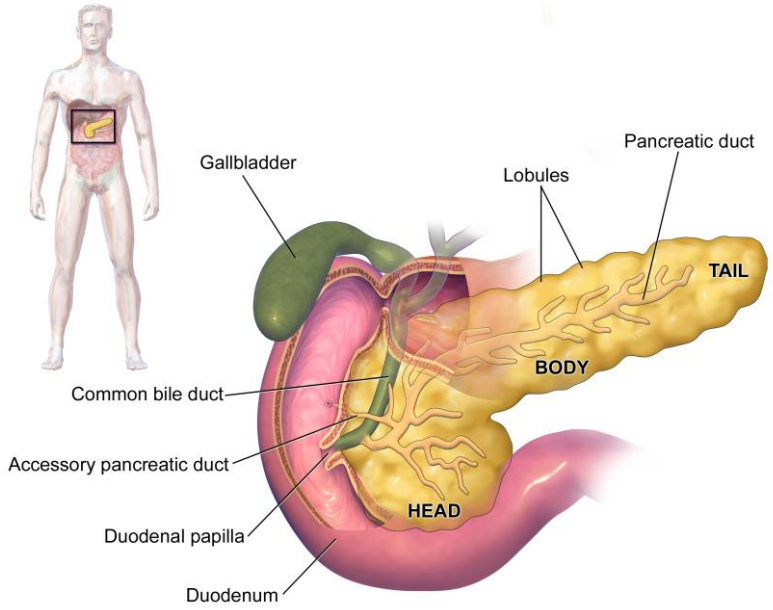
1. **In the absence of light**:
- cGMP is high due to GC function, and ion channels get opened to incorporate ions (e.g., Na+).
- Consequence: Mbr. depolarization which is sensed by brain neurons as dark.

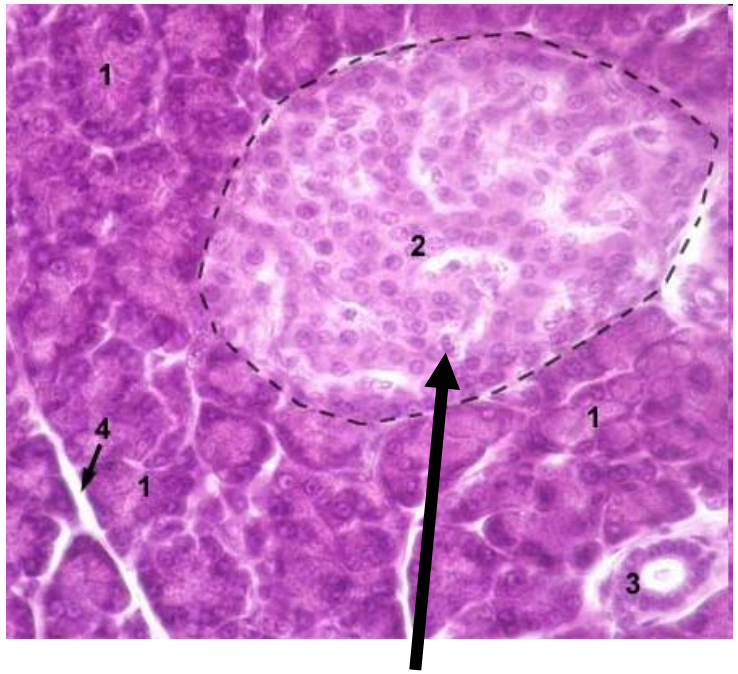
2. In the presence of light
- Light stimulates Rhodopsin, a GPCR, in retinal rods to activate G protein and, then, phosphodiesterase.
- Consequence: reduces cGMP, closes ion channels, and yields mbr. hyperpolarization sensed as light .



cGMP Signaling stops → Brain detects light increase

Example of GPCR signaling: the glucagon receptor





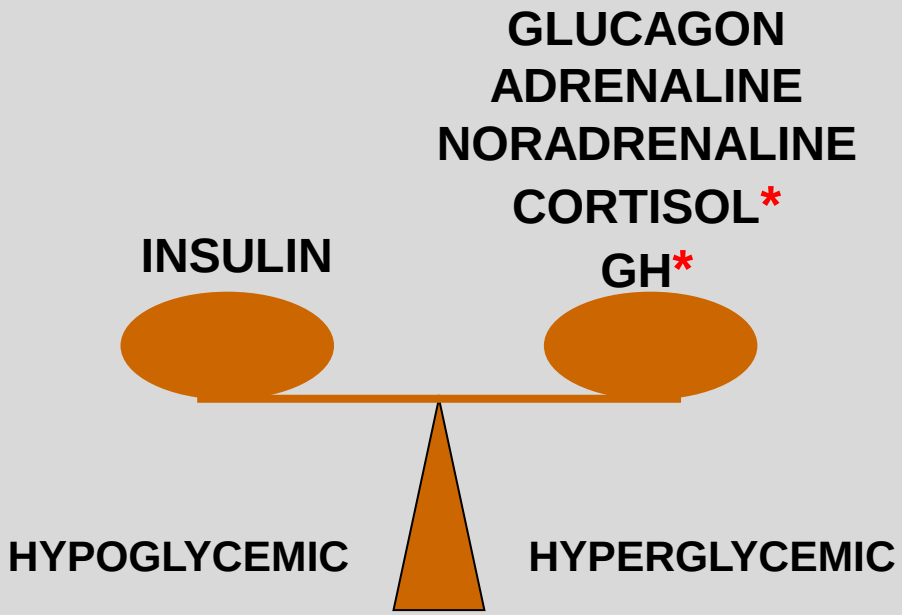
Islet of Langerhans



b cells: insulin

a cells: glucagon

Hormones that control glycemia



* Slow effect

Normal levels of glycemia:
70-110 mg/dL (fasting)

↓ **Levels: Hypoglycemia**

↑ **Levels: Hyperglycemia**

BRAIN, RED BLOOD CELLS can only use **glucose (not lipids or proteins)**

Alteration by GPCRs in glucose regulation

DIABETES: chronic disease in which blood glucose levels cannot be regulated (lack of insulin or insuline resistance)

Signs of hypoglycemia

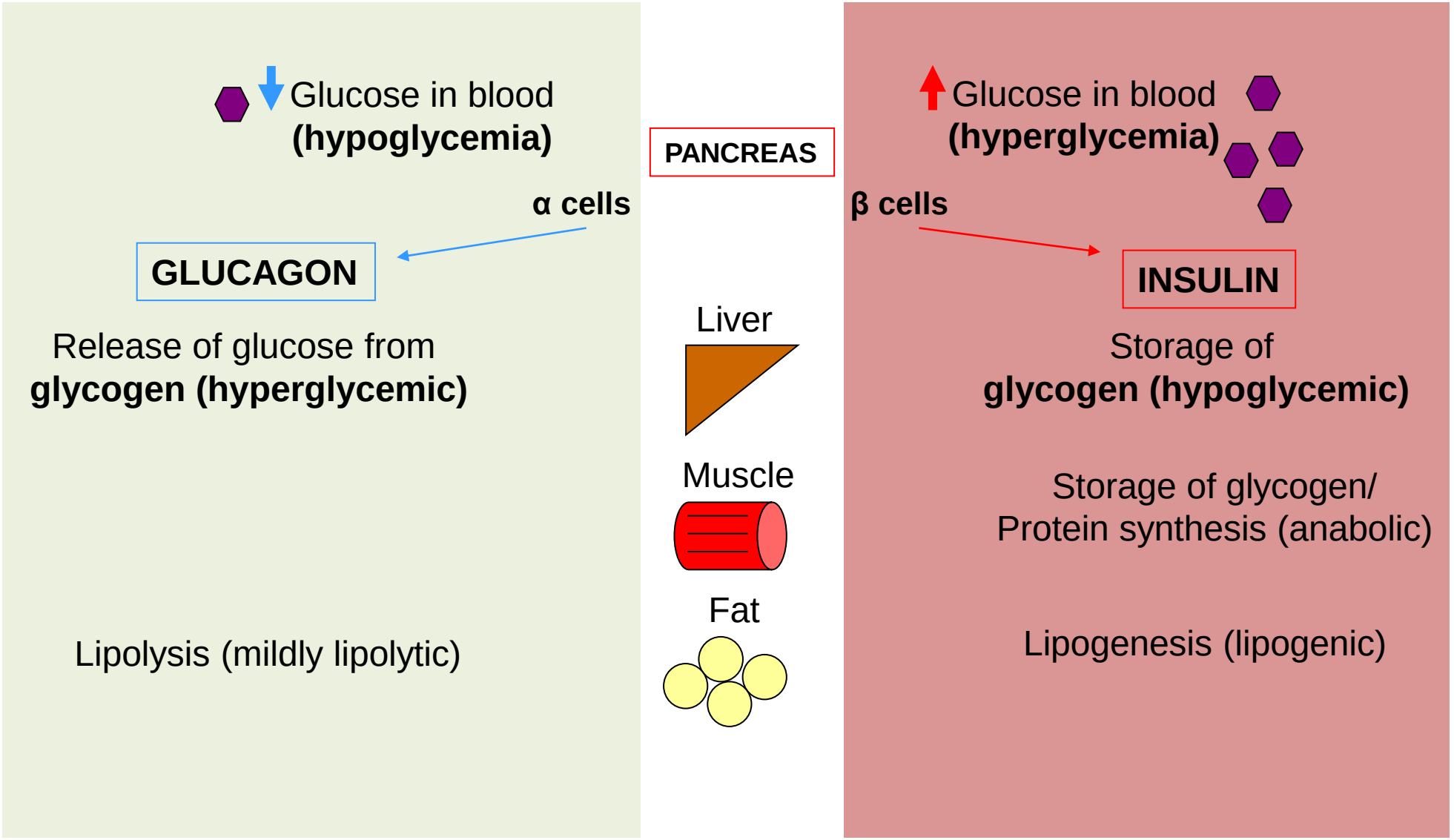
Shakiness, dizziness, sweating, hunger, fast heartbeat, confusion, anxiety, headache

Difficulty at speaking, double vision, convulsions or seizures, unconsciousness, death (rarely)

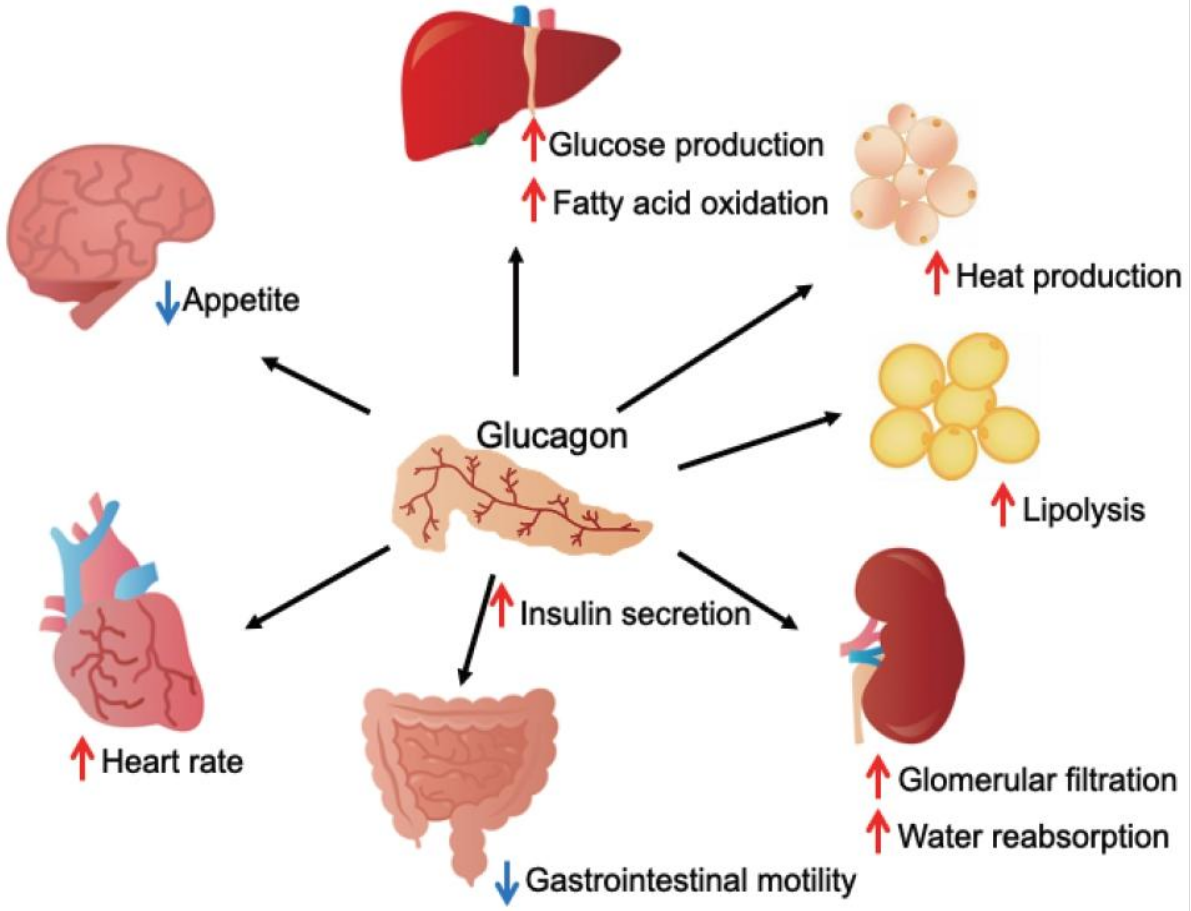
Causes of hypoglycemia in diabetic persons

Too much insulin
Not eating enough/ skipping meals
Increasing exercise without eating more or adjusting the amount of insulin
Excessive alcohol drinking

Regulation of glycemia by insulin/glucagon



Glucagon targets multiple tissues



Glucagon signal involves multiple enzymes

